

05 — Experiment Program & Research Story

Turn model training into evidence and a credible portfolio case study

Build guide for the Context-Aware Online Distress Severity Detection project.

1. Experiment Discipline

Every claimed improvement should come from a controlled comparison.

Change one major factor at a time when possible.

Use the same data split and evaluation protocol when comparing models.

Record configuration, checkpoint, metrics, runtime, and observations for every run.

2. Recommended Experiment Ladder

```
E0 – Dataset and class-distribution analysis
E1 – TF-IDF + Logistic Regression
E2 – RoBERTa with ordinary cross entropy
E3 – RoBERTa + class weighting
E4 – RoBERTa + focal loss
E5 – RoBERTa + weighted focal loss
E6 – Domain-adapted transformer
E7 – Ordinal classifier/objective
E8 – Calibration / temperature scaling
E9 – Attention + attribution analysis
```

3. What Each Experiment Should Answer

E1 answers: how far can classical NLP go?

E2 answers: does contextual representation help?

E3/E4 answer: does changing the loss improve minority/severe-class performance?

E6 answers: does domain adaptation help online language?

E7 answers: does respecting the severity order help?

E8 answers: can confidence be made more reliable?

E9 answers: what patterns and failure modes does the model appear to use?

4. Error Analysis Workflow

```
Test predictions
↓
Separate TP / FP / FN by class
↓
Sort severe-class errors first
↓
Manually inspect samples
↓
Assign error categories
↓
Count categories
↓
Form a hypothesis
↓
Design an experiment or preprocessing change
↓
Retrain / reevaluate
```

5. Examples of Errors to Study

Figurative: 'this exam is killing me' may be ordinary frustration.

Sarcasm: literal sentiment can disagree with intended meaning.

Implicit distress: serious distress may appear without explicit crisis vocabulary.

Context dependence: a single post can be ambiguous without surrounding context.

Annotation disagreement: human labelers may reasonably disagree about severity.

6. What Would Count as a Strong Finding?

A strong finding is not merely 'RoBERTa scored higher.'

A stronger finding would be something like: contextual transformer representations substantially reduced figurative-language false positives compared with TF-IDF, while a severe-class-aware loss improved severe recall with a measured precision trade-off.

Only make such statements after the experiment actually demonstrates them.

7. Research / Portfolio Narrative

The final case study should tell a causal engineering story:

Problem → data risks → baseline failure → transformer improvement → loss-function experiment → ordinal experiment → calibration → error analysis → final model selection.

This demonstrates ML engineering judgment rather than just library usage.

8. Lessons and Architectural Takeaways

Contextual transformer architectures are well suited to nuanced informal language.

Ordinal or severity-aware objectives can better represent ordered risk categories than a naive binary target.

Privacy and ethical sanitization belong at the start of the pipeline, not as a final documentation paragraph.

Confidence must be evaluated rather than blindly trusting softmax scores.

Model errors are research evidence and should drive the next experiment.